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Towards Context-Aware Clinical Decision Support with Retrieval-Augmented Generation

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Abstract

This research explores the Retrieval-Augmented Generation (RAG) method in the application of the Large Language Model to help Indonesian doctors in clinical decision making. By using the Clinical Practice Guidelines implemented by the Ministry of Health of the Republic of Indonesia, the answers generated by the model are expected to be in accordance with existing standards so that doctors are able to make more precise decisions. Two models are implemented with RAG in this research, GPT-4o mini by OpenAI and Claude 3.5 Haiku by Anthropic. From the evaluation that has been done in this research, it is proven that the models implemented with RAG are better in terms of the accuracy of disease diagnoses for the symptoms given to conformity with the Clinical Practice Guidelines.

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1. Main text

Artificial Intelligence has greatly revolutionized a number of industries, and probably one of the most exciting yet relatively unexploited domains is healthcare [1]. The potential of AI to revolutionize clinical decision-making and diagnostic accuracy is quite huge, advancing tools to medical professionals for making prudent decisions more rapidly and with far greater precision [2]. While AI has increasingly crept into the healthcare workflows from automatic

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diagnostic systems to predictive analytics over the patient outcomes in developed countries, this is far from happening in developing countries like Indonesia, for whom the minimum use of AI in healthcare is still a long-standing story [3]. This gap arises owing to a multitude of factors relating to data privacy, model reliability, and lack of relevant infrastructures designed for local health [3]. With the large population and dispersed health care in Indonesia, it faces special challenges that hamper the general diffusion of AI into clinical applications [1].

Resource limitations, a maldistribution of medical expertise, and a need for locally tailored solutions often burden the Indonesian healthcare system. In addition, there is skepticism among medical professionals about the accuracy and safety of AI-driven tools, especially when these are deployed without direct human oversight [4]. These chal- lenges, therefore, have to be overcome with novel thinking not only to remove these barriers but also unleash the transformative power of AI [5]. The paper proposes a novel methodology in addressing the challenges by using Retrieval-Augmented Generation, or RAG, a class of state-of-the-art AI method that integrate generative capabilities with retrieval-based systems.

The RAG model was developed to offer accessible diagnostics, quick, and capable of being performed by a general practitioner or a radiologist. The model is aimed at helping doctors in speedy diagnosis; thus, there can be improvement in patients' condition, and the efficiency in treatment will be enhanced while also helping reduce the stress experienced in the operations in Indonesian Healthcare. This, therefore, calls for a critical understanding of how sophisticated AI techniques such as RAG could be adapted to the peculiar challenges of developing countries to pave the way toward an equitable and effective global healthcare system.

2. Related Works

The use of Artificial Intelligence (AI) technology in the health sector, especially for clinical decision-making, is rarely found in Indonesia. This is due to its questionable accuracy, lack of safety when implemented in real-world scenarios, also the fear of leaking patient privacy data [6] [7]. A study states that the use of AI in direct practice cannot be recommended when used independently without supervision from a doctor, although it performs quite well when working on theoretical medical exam questions [8].

Even so, there have been quite a few groups or individuals who have tried to create AI models specialized for medical tasks. One of them is MediTron, a set of open-source large language models (LLMs) designed specifically for medical use. MediTron is designed to improve lay people's access to medical knowledge as well as support medical decision- making [9]. There are two main variants of this LLM, MediTron-7B and MediTron-70B, which are distinguished by the number of parameters, 7 billion and 70 billion respectively. The performance of MediTron-70B, which is built based on Meta's Llama-2 model, is proven to be able to surpass models in its class such as Clinical-Camel-70B, Med42-70B, and even its own base model, Llama-2-70B [9].

To address the issue of inaccuracy and outdated of answers generated by LLM, a technique called Retrieval-Augmented Generation (RAG), which makes LLM capable of reading a PDF file, has been invented [10]. RAG accommodates several steps required by LLM in order to achieve the expected functionality, including external data search which is used to obtain information that has the same context [11]. This step is then used to generate the answers needed by the user. On the other hand, the PDF format files that will be used as sources, need to be extracted first and converted into vectors through a process called embedding [12]. This needs to be done to make it easier for the computer to retrieve the information needed to ensure the accuracy of the LLM.

In Indonesia, there is one Large Language Model trained specifically for medical purposes in Indonesian language. Kesehatan-7B-v0.1, an open-source models developed by Arkan Bima [13]. This model was developed with the Mistral-7B-v0.1 base model by Mistral AI. Unfortunately, there is no complete documentation about this model so additional testing is needed to be able to know the performance of this LLM thoroughly.

The Ministry of Health of the Republic of Indonesia as the formulator, setter, and implementer of policies in the field of public health in Indonesia [14], issued a document containing a decree of the Minister of Health of the Republic of Indonesia entitled "Clinical Practice Guidelines for Doctors in First Level Health Care Facilities" [15]. The decree document aims to harmonize standard operating procedures for medical services at first-level health care facilities. The document contains various names of diseases, symptoms of the disease (subjective), the results of physical examinations and simple supporting examinations of the patient (objective), diagnosis establishment (assessment), comprehensive management plans, treatment support equipment, to prognosis or prediction of the possibility of the

patient's treatment outcome [15]. Data from these diseases from the document will later be used as a valid reference by the Large Language Model to answer questions or prompts given by the user by utilizing the RAG method.

3. Methodology

3.1. Pre-processing Data

Two types of datasets were used in this research. The first datasets was extracted from the guidelines outlined in the Keputusan Menteri Kesehatan Republik Indonesia Nomor HK.01.07/MENKES/1186/2022 tentang Panduan Praktik Klinis bagi Dokter di Fasilitas Pelayanan Kesehatan Tingkat Pertama (Decree of the Minister of Health of the Republic of Indonesia Number HK.01.07/MENKES/1186/2022 on Clinical Practice Guidelines for Doctors in Primary Health- care Facilities). This document, published by the Indonesian Ministry of Health, provides detailed clinical guidelines that served as a reliable resource for creating a text corpus for RAG. This document was obtained in PDF format from the official government publication. The second datasets contains over 100 data was obtained in DOCX format by the experienced doctor from patient's consultation and was used for model evaluations. The sample data and its explanations of each column can be seen from Table 1. Given the complexity and variability of the input, extensive pre-processing was conducted to standardize the data for model training. This included:

3.1.1. Splitting PDF

The first dataset was split into 15 separate PDFs and categorized based on groups of illnesses.

3.1.2. Text Cleaning

The second dataset, consisting of over 100 DOCX files, was extracted and converted into CSV format then categorized into 22 columns. Medical abbreviations and shorthand notations were expanded or standardized based on domain knowledge and through consultations with medical professionals to ensure accurate interpretation by the model.

3.1.3. Handling Missing Data

Any unknown or missing information in the symptom description was systematically replaced with "tidak terdata" to create consistency across records in the second dataset.

Table 1. Consultation Dataset Table

Attribute	Description	Sample Value/Range
keluhan utama	The symptoms that describes by the patients	pasien datang dengan keluhan nyeri ulu hati (+) terasa menyesak sejak 2 hari smrs. mual (+) muntah (-)
jalannya penyakit	Progression of an illness	sudah minum obat belum ada perubahan
Pemeriksaan penunjang	Follow-up examinations carried out by doctors to diagnose certain diseases or in- juries	td: 130/100 hr: 90 21 NTE (+) vas scale: 8/10
hasil laboratorium	Laboratory results	Terlampir, -
diagnosa utama	Diagnosis of disease by a doctor	Dyspepsia, Fever, unspecified, Asthma
kode icd utama	The ICD-10 code for its disease	K30, J45, R50
kondisi pulang	The patient's condition after examined by the doctor	Hidup
obat waktu pulang	Medicines that doctors give to their pa- tients	Inj Ketorolac 1 ampul Inj Ranitidine 1 ampul Fasidol 3x1 tab Asamefenamat 3X1 tab

3.2. Pre-processing Data

For this research, the model provided by OpenAI is used for the vectorization or embedding process. The OpenAI Embeddings model converts textual content into a vector representation that captures not only the characters contained

within, but also includes the semantic meaning of the input words, phrases, and sentences. As of December 2024, OpenAI provides three embedding models that can be used, namely text-embedding-ada-002 and two newer models, text-embedding-3-small and text-embedding-3-large. According to OpenAI [16], these two new models released early 2024 are said to offer a significant performance upgrade over their predecessor with more efficient resource usage. Moreover, for the text-embedding-3-small model used in this research, OpenAI states that it is five times cheaper than the older text-embedding-ada-002.

The result of the embedding process, which is a vector, will need to be stored in a Vector Database. This research uses DataStax's Astra DB cloud vector database service. Considerations such as convenience, cost-effectiveness, and performance offered by Astra DB make it the preferred solution for this work. The feature that allows users to integrate directly with the deployment platform used in this research, Vercel, greatly facilitates and speeds up the development process [17].

The embedding process begins by parsing the Clinical Practice Guideline PDF files from the Ministry of Health of the Republic of Indonesia that have been separated based on the available disease categories into raw text. Then the text goes through a cleaning process to remove whitespaces and special characters, such as bullet points or tables, contained in these files to maintain text consistency. The text that has been parsed and processed will be separated into chunks of text with the number of characters per chunk ranging from 1000-1200 characters.

The next step is to embed those chunks of characters by calling the OpenAI Embeddings API. The output of this embedding will be a JSON similar to the one below:


```
{
      " object ":     " l i s t " ,
     " data ":    [
      {
            " object ":       "embedding " ,
           " index ":    0 ,
            "embedding ":     [
             −0.006929823910992422 ,
             −0.005384203842911228 ,
               . . .     ( omitted     for     spacing )
             −4.542183216378452324 ,
             −0.024028142110502e−03
          ] ,
      }
   ] ,
     "model ":        " text −embedding −3−small " ,
     " usage ":    {
          " prompt tokens ":       3 ,
       " t o t a-l t o k e n s ":   3
   }
}
```


The key "embeddings" and the value inside which contains several floats inside a list is the vector representation of the text that has been previously entered.

The final step is to insert the result of the embedding model into the Astra DB vector database. Several steps are required here, including creating an account, initializing the database instance, and creating an API key to access the database. After that, in the database that has been created, a table is needed to store the results of the vector representation that has been generated. An explanation of the columns in this table will be explained in Table 2.

For a better understanding of the whole embedding and database storing process used in this research, refer to Fig. 1. The diagram explains the process that has been carried out visually.

3.3. Base Model and RAG

Two common LLM models have been implemented with the RAG method: GPT-4o mini by OpenAI and Claude 3.5 Haiku by Anthropic. These two models were also given specific instructions to answer medical questions and

provide complete information based on the guidelines provided by Indonesia's Minister of Health to ensure that the answers generated by the LLM are in accordance with the standards and regulations set by the government. Table 3 depicted the comparison table for the specification and pricing of both base model that we used.

Table 2. Vector Database Table

Attribute	Description	Sample Value/Range
id	Unique auto-incrementing primary key to identify each record	1, 2, 3, ...
document id	Identifier for the source document from which the chunk of text was extracted	0 PPK UMUM.pdf, 7 NEUROLOGI.pdf
chunk id	Unique identifier for the specific text chunk within a document	0 PPK UMUM.pdf _chunk 0
text	The actual extracted text content from the PDF chunk	b. Usia 6-11 bulan 100.000 IU/hari...
category	Classification or label for the text chunk (e.g., disease category)	Penyakit Umum, Kardiovaskuler
embedding	Binary representation of the semantic embedding vector for the text chunk	Array of floats stored as bytes

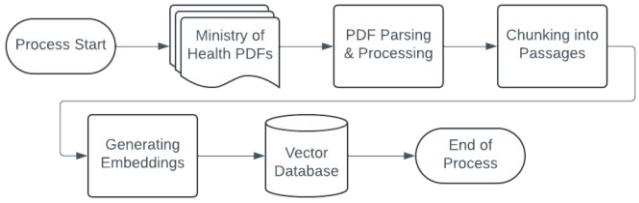


Fig. 1. Process of the Document Vectorization

Table 3. Specification Comparison between Claude 3.5 Haiku and GPT-4o Mini

Feature	**Claude 3.5 Haiku**	**GPT-4o Mini**
Input Context Window	200K tokens	128K tokens
Maximum Output Tokens	8,192 tokens	16.4K tokens
Release Date	November 4, 2024	July 18, 2024
Knowledge Cut-off Date	July 2024	October 2023
Input Cost	$0.80/1M Tokens	$0.15/1M Tokens
Output Cost	$4.00/1M Tokens	$0.60/1M Tokens

In the implementation, we utilized pre-embedded documents, conducted semantic search, and carefully structured the prompts. These steps are taken so that the LLM can present accurate, detailed, and contextualized answers according to the user's needs. All the details of each step in the RAG implementation can be seen in Fig. 2 and explained below:

User Query Input. At this stage, the query that can be in the form of questions and such from the user will be parsed to the backend of the web app to be processed in the next step.

API Request Handling. The query from the user is received by the backend to search for the most relevant information from the vector database.

Query Embedding. The backend converts the user's query from text to numerical embedding (a vector), allowing the machine to perform semantic search and capture its meaning. It is necessary to ensure that the embedding model used when embedding PDF files is the same as the one used in this step to ensure the similarity of the vectorization calculation, which in this case is OpenAI's text-embedding-3-small.

Vector Similarity Search and Ranking. Using the results of query embedding, the backend searches from the set of document chunks that have been embedded and stored in the vector database to then rank the chunks that are most

similar or semantically related to the user's query. Similarity between vectors is calculated using the Cosine Similarity formula where two vectors are calculated as dot product and normalized by their magnitude.

Prompt Construction. After getting the context of the query and the retrieved chunk, the backend will construct a complete prompt that will be passed to the LLM. In the prompt, instructions have been set up for the LLM to follow in answering the question, along with the retrieved chunk to be used by the LLM as the basis for its response. Some restrictions, rules, and disclaimers such as "This answer is not a substitute for professional medical advice" are added to the model instructions.

Model Text Generation. With the previously constructed prompt, the backend will send the prompt to the LLM via API. Then the model will read the prompt and instructions and generate a response that answers the user's question according to the instructions and follows the context and details of the retrieved chunks.

Streaming the Answer Back. While the model generates the answer, the backend of the web app will stream the generated text to the user, allowing the user to see the answer being generated by the model, until it is fully generated.

Applying the RAG process to well-established models, such as GPT-4o mini and Claude 3.5 Haiku, will improve the performance of the model and ensure that the model works according to the standards that have been implemented in the healthcare field in Indonesia.

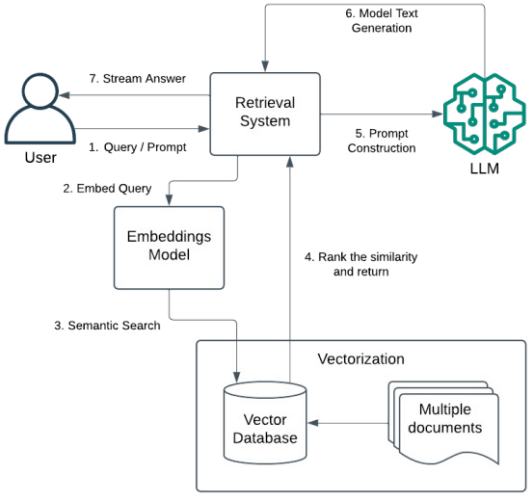


Fig. 2. Retrieval-Augmented Generation Workflow

3.4. Evaluation

For evaluation, different sets of metrics have been combined together to identify the performance of both the proposed RAG and the standard language models. These metrics would grant a full-scale analysis regarding quality, accuracy, and semantic similarity of model outputs compared to the expected diagnoses. The evaluation is done on three models namely GPT-4o mini, Claude 3.5 Haiku, and Kesehatan-7B-v0.1, and 2 of those models, GPT-4o mini and Claude 3.5 Haiku applied with RAG. Kesehatan-7B-v0.1 is compared here as a representative of one of the Indonesian opensource models specifically trained using health data. These models will be evaluated and compared with these metrics:

Precision, Recall, and F1-Score. These metrics are derived from a confusion matrix consisting of:

- True Positive (TP): the model predicts the expected diagnosis correctly.
- False Positive (FP): the model predicts a wrong diagnosis.
- False Negative (FN): the model does not provide a diagnosis when an expected diagnosis exists.

METEOR Score. METEOR (Metric for Evaluation of Translation with Explicit Ordering) score assesses lexical and semantic similarity between the model-generated outputs and the texts used as the reference. The score uses exact matches, synonyms, stems, alignment, and order [18].

Precision, Recall, and F1-Score. These metrics are derived from a confusion matrix consisting of:

- Precision (***P***): Proportion of matched unigrams in the candidate output relative to the total unigrams in the candidate
- Recall (***R***): Proportion of matched unigrams relative to the total unigrams in the reference
- Harmonic Mean (F_{mean}): Balances Precision and Recall with a weighted factor
- Penalty: Penalizes the ”chunkiness” or misalignment of matched unigrams
- METEOR Score:

$$METEOR = F_{mean} \cdot (1 - Penalty) \quad (1)$$

The purpose of METEOR is to evaluate the quality of generated outputs based on lexical matches, semantic meaning, and fluency. Ensuring the model implication to the standard set by the Indonesian Ministry of Health.

Cosine Similarity. Similar with METEOR evaluation, Cosine Similarity measures the semantic alignment between the model-generated output and reference texts. However, Cosine Similarity represents both texts as embeddings in a high-dimensional vector space [19]. The same embedding model, text-embedding-3-small, used in this evaluation process.

To handle multiple retrieved references, the embeddings of retrieved passages are averaged and compared to the model output. The purpose of Cosine Similarity is to capture the semantic closeness between generated outputs and references, complementing the word-level analysis of METEOR [20]. With those three metrics calculations, all models can be assessed for performance and compliance with the standards that have been implemented by the Indonesian government. As for the evaluation procedure, the following steps will be taken:

1. Data Preparation:
    - A set of 30 medical prompts is used, each with a corresponding expected diagnosis.
    - Outputs from all five models are collected for comparison.
2. Model Execution: Each model generates a diagnosis based on the given prompts.
3. Metrics Calculation:
    - Precision, Recall, and F1-Score are calculated based on exact matches with expected diagnoses. A sample prompt containing the symptoms of the patient is given to each model with the expectation of the model being able to identify the diagnosis. For example, ”Pasien terbakar petasan di tangan kiri. Pasien didiagnosis penyakit apa?” The precise diagnosis would be a first-degree burn based on the data which was given through doctor’s consultations.
    - METEOR Score evaluates lexical and semantic overlap. A sample prompt would be used in order to check similarity. The prompt will instruct the LLM to explain fully about a specific disease. For example, ”pasien kemungkinan terkena Rinitis Vasomotor, berikan informasi lengkap mengenai Rinitis Vasomotor.” The LLM would then proceed to provide everything they know about it.
    - Cosine Similarity measures semantic alignment using vector embeddings. It uses the same prompt used for METEOR score and it compares the result of the response with the proper medical data.
4. Comparative Analysis: The results are compared across all five models to identify their strengths and weaknesses.

3.5. Case Demonstration and System Behavior

In order to illustrate how the Decision Support System (DSS) is applied to actual cases in real-life scenarios, we integrated the chatbot with actual patient information from the consultation dataset, including the symptoms, disease development, and test results.

As an example, a patient experiencing diarrhea, epigastric pain, and nausea was accurately diagnosed with Dyspepsia (ICD-10: K30). The model searched for appropriate guidelines and proposed corresponding treatments, including ranitidine and antacids, in line with physician comments. In another case, a patient developed rashes on the hands and feet along with fever. Herpes Zoster (ICD-10: B02) was recognized by the chatbot, and standard antiviral medication such as acyclovir was recommended. These examples demonstrate that the system can match different symptom patterns and retrieve medically accurate answers from the official clinical guidelines.

4. Results and Discussion

4.1. Model Evaluation

30 medical prompts were collected for measuring the performance of various RAG models, all of which needed comprehensive information with regard to certain diseases. The performance of each model are measured by METEOR score, Cosine Similarity, Precision, Recall, and F1 Score. The details of the results of all models can be seen from Table 4 and Table 5. Both the RAG-based models, GPT 4o Mini + RAG and Claude 3.5 Haiku + RAG, had higher METEOR scores of 0.2081 and 0.2186, respectively, compared to their non-RAG counterparts, hence aligning more closely with the reference medical texts.

The highest semantic similarity was from GPT 4o Mini + RAG at 0.7004, while Claude 3.5 Haiku + RAG reached a close second with 0.6902. The rest of the non-RAG models are much lower in value, confirming the semantic relevance given by RAG. The RAG models have perfect recall at 1.0000, and with better precision scores; hence, the highest F1 score for Claude 3.5 Haiku + RAG was 0.8400. In contrast, non-RAG models gave precisions and F1 scores lower than the rest, while Kesehatan could not provide valid outputs based on the given prompt.

Table 4. METEOR and Cosine Similarity Scores Across Models

Model	METEOR Score	Cosine Similarity
GPT-4o mini + RAG	0.2081	0.7004
GPT-4o mini	0.1263	0.6211
Claude 3.5 Haiku + RAG	0.2186	0.6902
Claude 3.5 Haiku	0.1160	0.5172
Kesehatan-7B-v0.1	0.0814	0.5665

Table 5. Precision, Recall, and F1 Score Across Models

Model	METEOR Score	Cosine Similarity
GPT-4o mini + RAG	0.8163	1.0000
GPT-4o mini	0.3103	1.0000
Claude 3.5 Haiku + RAG	0.7241	1.0000
Claude 3.5 Haiku	0.5172	1.0000
Kesehatan-7B-v0.1	N/A	N/A

4.2. Deployment

This research is deployed using several tools starting from the frontend to the hosting process. We use Next.js framework to develop both frontend and backend. The frontend in Next.js is responsible for providing a user interface using TypeScript. While the backend serves to call a set of APIs and perform processes behind the scenes of the web app, such as calling OpenAI and Anthropic APIs for Embeddings and Generative Models, retrieving vector data from Astra DB, and streaming model-generated answers to users. Then for the hosting, we use Vercel, a cloud platform that makes it easy for developers to build, host, and manage web apps. Vercel was developed by the same team that created Next.js. This allows us to easily and quickly integrate the web app we have created with the hosting platform.

Another consideration for using Vercel was the ease of collaborating with the team. Vercel can be integrated with GitHub, a platform for developers to collaborate. Every time we add something or edit something from GitHub and push it to the repository, Vercel will automatically build and deploy the web app. Also, with Vercel, we don’t have to worry about hardware settings like on conventional deployment platforms.

4.3. Limitations

This research, however, is subject to several limitations. Limited development costs are one of them, where we have to find the cheapest solution in order to carry out this research within the budget range. This limitation resulted in some shortcomings in terms of web app performance. For example, Astra DB does provide free services for users but, the region used to store the vector data is located in the East US region. This makes users who access the web app outside of the US have to experience longer latency to get answers from the model. On the hosting side, using Vercel with the Hobby plan has limited resources. Resources such as bandwidth, data transfer, and requests are limited per month. Therefore, the web app can suddenly become inaccessible due to exhausted resources. For small-scale use cases, this can still be tolerated, but to scale up the use case, additional costs are required to accommodate the large amount of incoming and outgoing traffic.

Another important limitation to mention is the limited computing power we have. The original idea for this research was to fine-tune the pretrained large language model so that it fits our needs completely. However, with the insufficient resources available to conduct the fine-tuning process given the size of the model and the dataset for training, we decided to change our method to RAG. This way, we cannot fully customize how the model responds as there are already some built-in instructions from the third-party model we use.

5. Conclusion and Future Work

This research demonstrates the potential of Retrieval Augmented Generation (RAG) to help enhance decision making processes within the Indonesian healthcare system. Using Large Language models such as GPT-4o, GPT-4o mini, and Claude 3.5 Haiku that integrate with RAG, our proposed solution can help the gap between theoretical advancements and practical applications. Notably, this method will help doctors especially Indonesian to make a clinical decision when diagnosing a patient's illness based on symptoms.

The evaluation metrics show the important impact of RAG based models compared to baseline implementations, particularly in terms of METEOR scores, cosine similarity, and F1 scores. The integration of Indonesian medical guidelines ensures the alignment with local healthcare standards, thereby helping doctors create informed decisions. Nevertheless, there are limitations such as latency, budget restrictions, and dependency on third party models. However, our proposed method presents a scalable foundation for future advancements.

Future developments will focus on addressing the current limitations of the system to ensure broader adoption and improved performance. Efforts like include exploring lightweight, locally runnable AI methods to reduce reliance on third party models and high compute power, which will make the solution more cost effective. Data privacy will also be a key priority, with methods such as encrypting user's queries and securing embedded RAG data to ensure privacy and build trust among healthcare professionals. Moreover, optimizing web app performance and infrastructure to minimize latency. These improvements will create a better system for clinical decision making in Indonesia.

Data Availability

Due to the sensitive nature of the questions asked in this study, survey respondents were assured raw data would remain confidential and would not be shared. All the codes used in this paper can be accessed through the following link: *https://github.com/aubertlenno/radiology-gpt.*

Author Contribution Statement

L.A Hartono, I.K.G.P. Widagda, G.N.R. Agastya: Conceptualization, Data curation, Formal Analysis, Methodology, Software, Visualization, Writing - Original draft preparation, Writing Reviewing and Editing. N.N. Qomariyah, F. Setiawan: Conceptualization, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

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